Sanando Banerjee

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PROFESSIONAL SUMMARY

Software Engineering undergraduate specializing in backend development and AI-powered systems. Experienced in building scalable FastAPI services, RAG-based recommendation pipelines, and real-time data visualization platforms. Strong foundation in data structures and cloud deployment (AWS/Azure).

EDUCATION

Meghnad Saha Institute of Technology — Kolkata, India
Bachelor of Technology (B. Tech) in Information Technology — August 2023- July 2027

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript
Frameworks & Libraries: LangChain, LangGraph, NextJS
Backend & APIs: FastAPI, MCP Server
Databases: MongoDB, ChromaDB, SQL Server
DevOps and Tools: Git, Docker, CI/CD, AWS, Azure

RELEVANT EXPERIENCE

- **bITsera – Tech Club of IT, MSIT** (*Web Development Lead)*
  Led and organized React workshops for 60+ students, covering state management, API integration and React hooks while emphasizing the importance of modular architecture and reusable component design.

PROJECT EXPERIENCE

Saturn-AI – Full-Stack AI Trading Intelligence Platform

- Engineered a full-stack Bitcoin intelligence platform that ingests live cryptocurrency news from **5 RSS feeds** every **10 minutes**, leveraging **Groq**, **FastAPI**, and **MongoDB Atlas** to generate AI-powered **BUY/SELL/HOLD** trading signals with confidence scoring.
- Designed a modular backend following **SOLID principles**, implementing **9+ REST APIs**, scheduled asynchronous pipelines, a streaming AI explanation service, and an **MCP server** exposing trading tools to Claude Desktop and other MCP-compatible AI agents.
- Developed a **React** dashboard delivering real-time sentiment feeds, Bitcoin price tracking via **CoinGecko**, streaming AI market analysis, and interactive visualizations through low-latency backend services.

DDoS Overwatch – Real-Time Cyberattack Visualization Platform

- Engineered real-time DDoS attack visualization dashboard processing **2+ attacks/second** across 14-country geographic network using Server-Sent Events and React, with centralized Zustand state management supporting **1,000+ concurrent events**.
- Built geospatial cybersecurity analytics dashboard with **real-time data processing pipeline**, sub-second UI responsiveness, and **1M-100M bps bandwidth** tracking metrics.
- Improves **situational awareness** in cybersecurity by making large-scale, real-time DDoS activity understandable and actionable when exposed to real-world datasets.